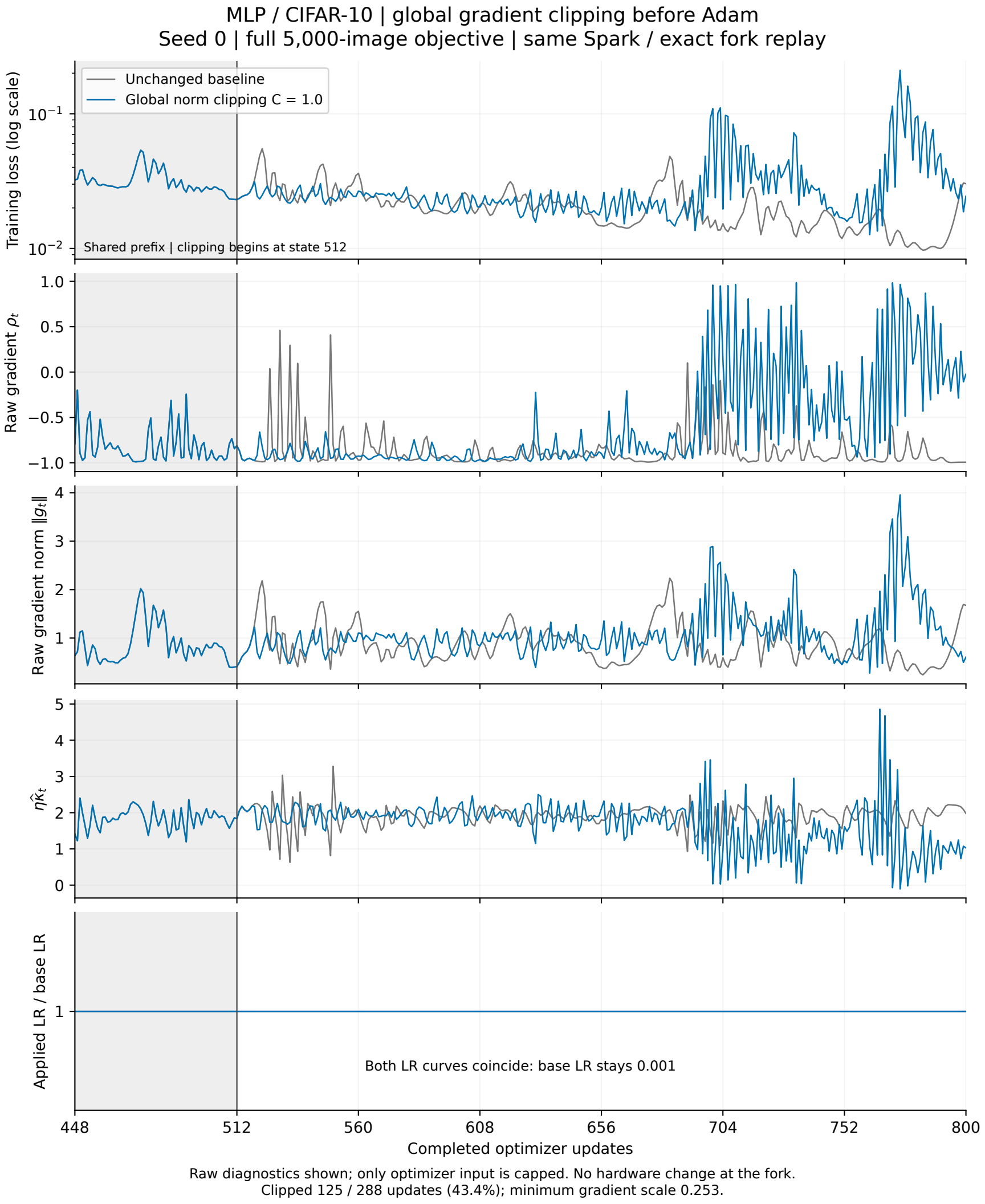

MLP / CIFAR-10 | global gradient clipping before Adam
Seed 0 | full 5,000-image objective | same Spark / exact fork replay
Unchanged baseline
Global norm clipping C = 1.0
Training loss (log scale)
Shared prefix | clipping begins at state 512
Raw gradient ρ_t
Raw gradient norm $\|g_t\|$
$\eta\hat{\kappa}_t$
Applied LR / base LR
Both LR curves coincide: base LR stays 0.001
Completed optimizer updates
Raw diagnostics shown; only optimizer input is capped. No hardware change at the fork.
Clipped 125 / 288 updates (43.4%); minimum gradient scale 0.253.